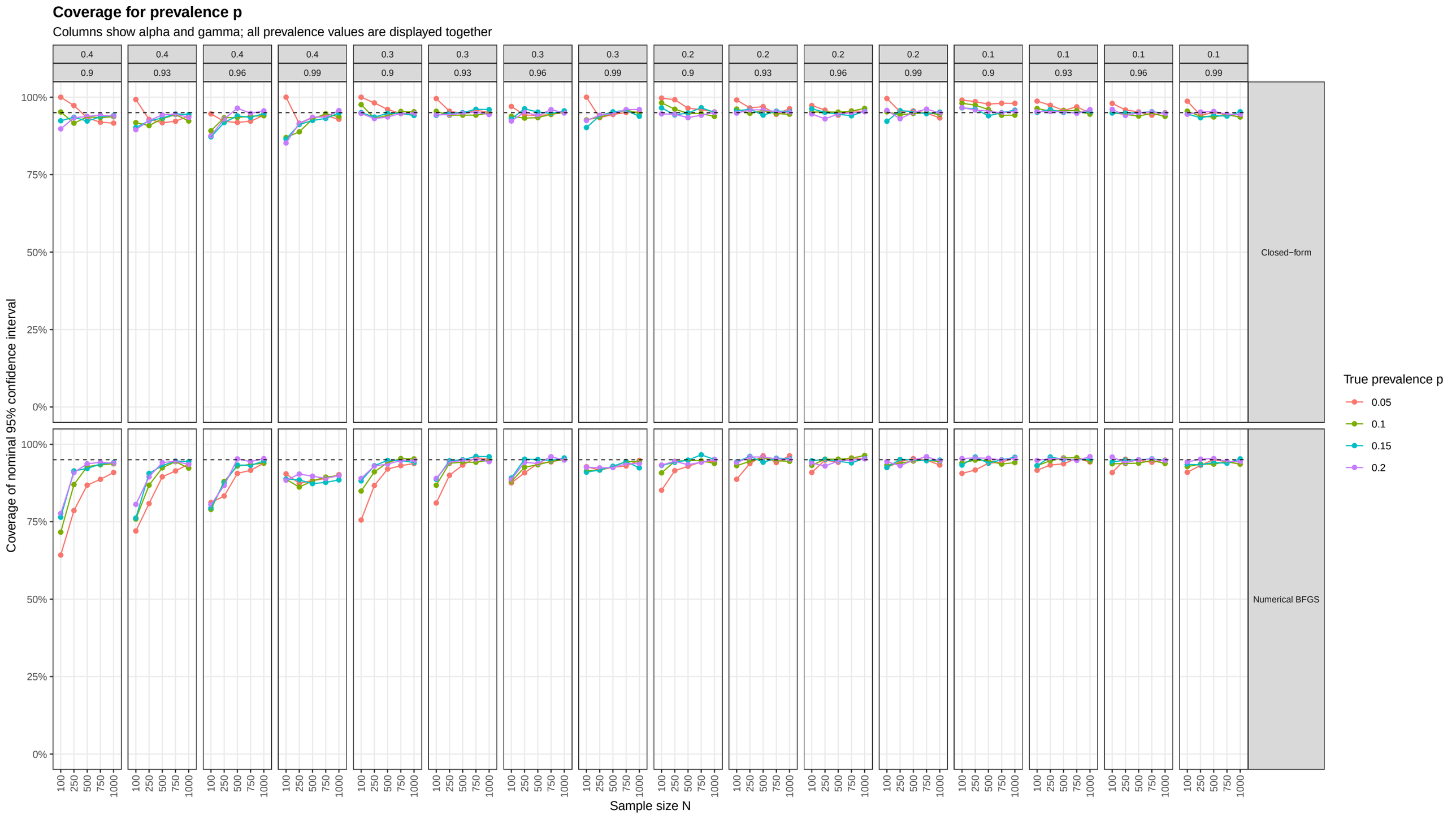
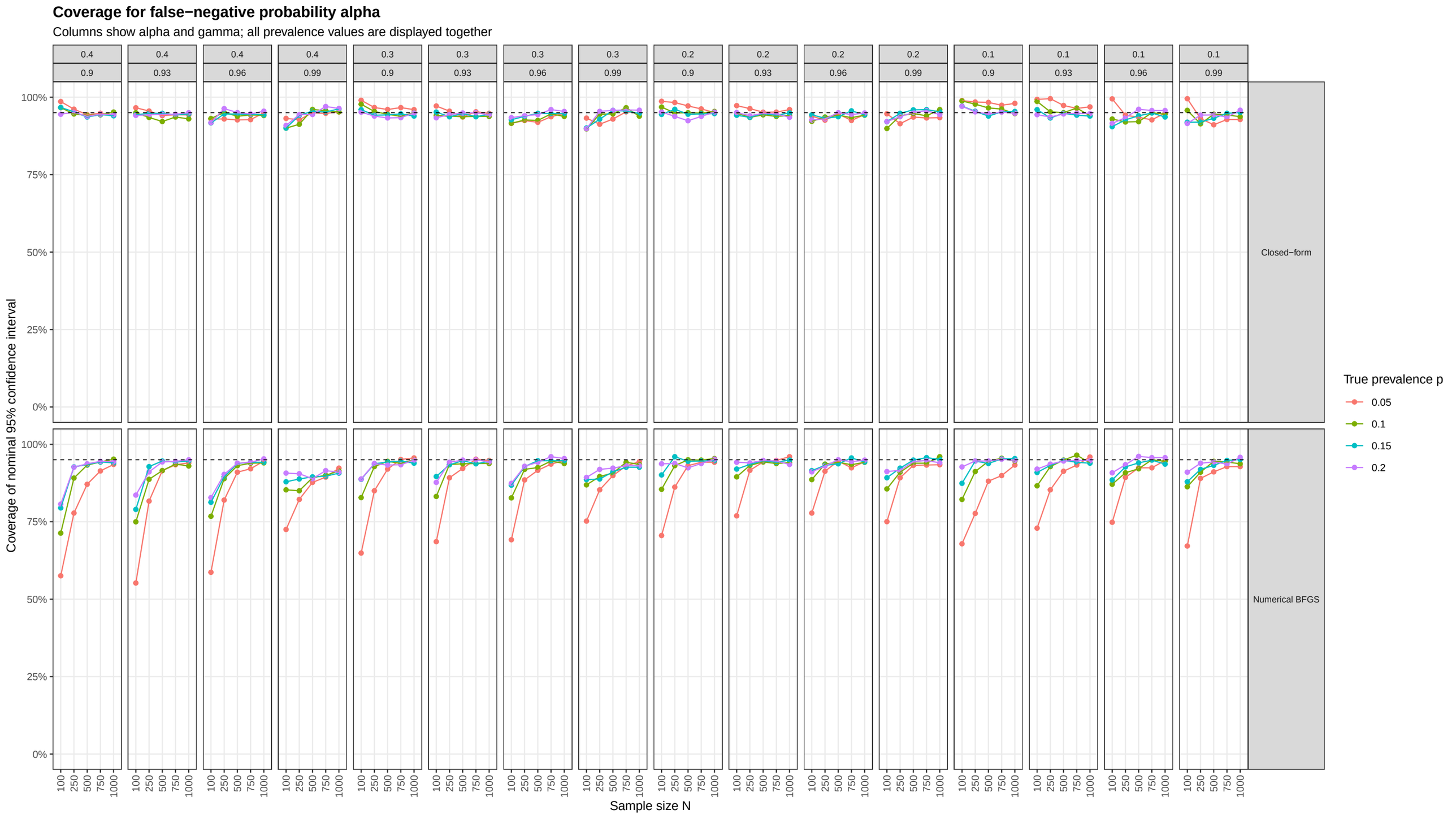
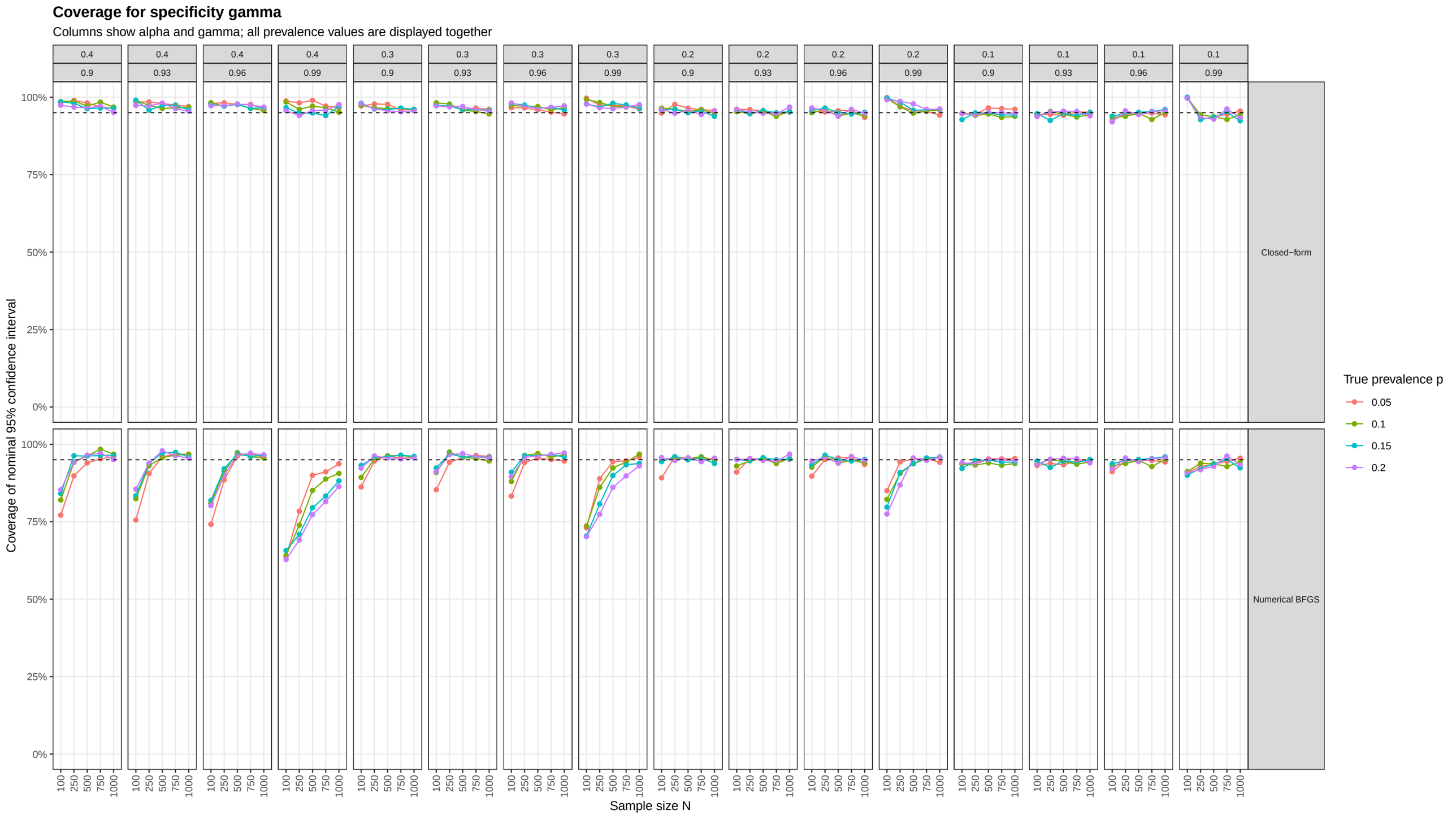

Coverage for prevalence p
Columns show alpha and gamma; all prevalence values are displayed together
0.4
0.9
0.4
0.93
0.4
0.96
0.4
0.99
0.3
0.9
0.3
0.93
0.3
0.96
0.3
0.99
0.2
0.9
0.2
0.93
0.2
0.96
0.2
0.99
0.1
0.9
0.1
0.93
0.1
0.96
0.1
0.99
Closed−form
Numerical BFGS
Coverage of nominal 95% confidence interval
100%
75%
50%
25%
0%
100
250
500
750
1000
Sample size N
True prevalence p
0.05
0.1
0.15
0.2


Coverage for false−negative probability alpha
Columns show alpha and gamma; all prevalence values are displayed together
0.4
0.9
0.4
0.93
0.4
0.96
0.4
0.99
0.3
0.9
0.3
0.93
0.3
0.96
0.3
0.99
0.2
0.9
0.2
0.93
0.2
0.96
0.2
0.99
0.1
0.9
0.1
0.93
0.1
0.96
0.1
0.99
Closed−form
Numerical BFGS
Coverage of nominal 95% confidence interval
100%
75%
50%
25%
0%
100
250
500
750
1000
Sample size N
True prevalence p
0.05
0.1
0.15
0.2


Coverage for specificity gamma
Columns show alpha and gamma; all prevalence values are displayed together
0.4
0.9
0.4
0.93
0.4
0.96
0.4
0.99
0.3
0.9
0.3
0.93
0.3
0.96
0.3
0.99
0.2
0.9
0.2
0.93
0.2
0.96
0.2
0.99
0.1
0.9
0.1
0.93
0.1
0.96
0.1
0.99
Closed−form
Numerical BFGS
Coverage of nominal 95% confidence interval
100%
75%
50%
25%
0%
100
250
500
750
1000
Sample size N
True prevalence p
0.05
0.1
0.15
0.2